

Assignment 5: 8-Bit LFSR

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1 Introduction

The Linear Feedback Shift Register (LFSR) is a critical block used for cyclic redundancy checks (CRC) and pseudo-random pattern generation. The designed block operates serially, taking one bit at a time and computing the CRC using a predefined feedback polynomial.

The module operates using a finite state machine that sequences through data input and CRC serialization phases.

2 Specifications

- **Clock and Reset:** The block is driven by a 10 MHz clock and uses an active-low asynchronous reset (`rst_n`).
- **Seed:** The shift register initializes to a seed value of `8'hD8` whenever reset is active.
- **Data Shifting:** Data is shifted serially into the `DATA` input port, LSB first. The shift is active only when the `ACTIVE` signal is high.
- **CRC Computation:** Feedback is generated using the logical expression `Feedback = LFSR[0]  $\oplus$  DATA`.
- **Valid Flag:** Once `ACTIVE` drops to low, the internal computed CRC is shifted out over 8 clock cycles, while the `Valid` output signal is driven high to denote valid data.

3 Simulation Waveforms & Analysis

The simulation verifies the operation of the LFSR over 10 random test cases utilizing a 10 MHz clock with a 50 ns low and 50 ns high duty cycle.

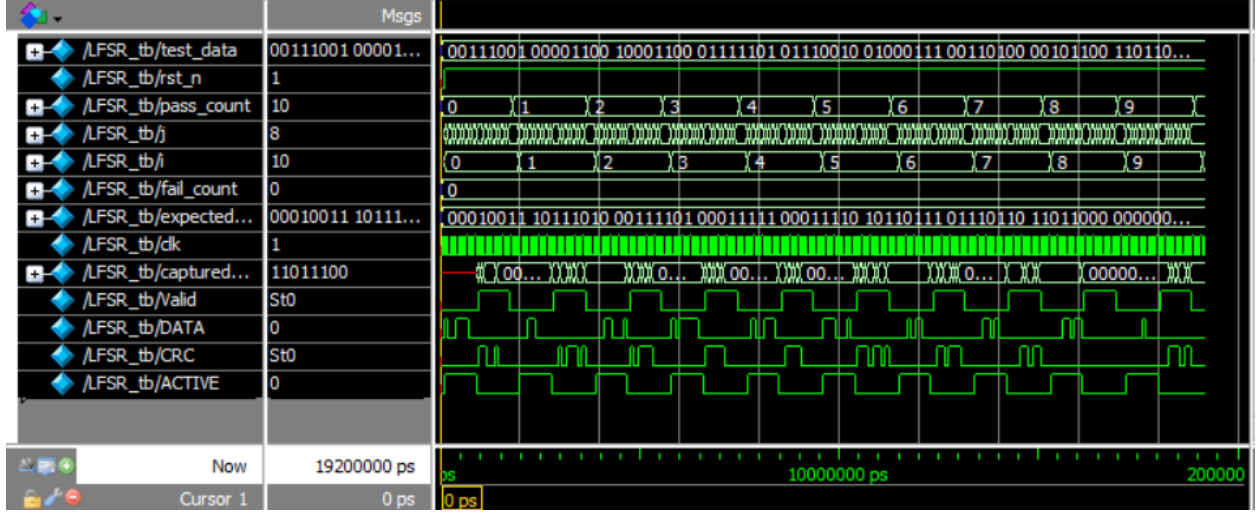


Figure 1: Full Simulation Waveforms of the 8-Bit LFSR

3.1 Waveform Analysis & Commentary

The simulation waveforms in Figure 1 demonstrate the functional correctness of the design across all 10 test cases. A detailed breakdown of the observations from the waveform is provided below:

- **Initial Reset (Time 0 to 100 ns):** At the start of the simulation, `rst_n` is held low. The internal shift register `lfsr_reg` is successfully initialized to the seed value `8'hD8`. The outputs `CRC` and `Valid` are registered to `1'b0`.
- **Test Case 1 Verification (Time 100 ns to 1900 ns):**
 - **Input Phase:** `ACTIVE` rises high, and the input byte `0x39` (`00111001` in binary) is shifted in LSB-first (`1, 0, 0, 1, 1, 1, 0, 0`) over 8 clock cycles of the 10 MHz clock.
 - **Output Phase:** Once `ACTIVE` falls low, the `Valid` flag immediately asserts high. Over the next 8 clock cycles, the computed CRC of `0x3C` (`00111100`) is shifted out LSB-first on the serial `CRC` line. As seen in the waveform, the `CRC` output is low for 2 cycles, high for 4 cycles, and low for 2 cycles, matching `0x3C` exactly.
 - **Success:** The testbench captures this serial stream, reconstructs the byte `0x3C`, compares it with `Expec_Out_h.txt`, and increments `pass_count` to 1.
- **Analysis at Cursor Position (Time 2.29 μ s):** The vertical yellow cursor is positioned at exactly 2,289,980 ps (2.29 μ s):
 - The testbench loop index is `i = 1`, denoting the second test case (Test Case 2).
 - The test input byte is `test_data[1] = 0x0C` (`00001100`).
 - The current bit counter is `j = 2`, which means the third bit (index 2) of `0x0C` is being shifted in. Since the byte is shifted LSB-first, the bits are 0 (bit 0), 0 (bit 1), and 1 (bit 2). Correspondingly, the `DATA` input line is high (`1'b1`) at the cursor.
 - `Valid` is low and `ACTIVE` is high, indicating the module is in the `SHIFT_DATA` state.
 - The `pass_count` is 1, reflecting the successful passage of Test Case 1.
- **Overall Results:** This sequence repeats for all 10 test cases. At the end of the simulation, all 10 test cases pass, and the final `pass_count` registers 10 with a `fail_count` of 0.

4 Simulation Output Logs

The simulation log validates each function's expected value (parsed from `Expec_Out_h.txt`) versus its actual captured CRC, resulting in 10 PASSED test cases and 0 FAILED out of 10.



```
Transcript
** Warning: (vsim-WLF-5000) WLF file currently in use: vsim.wlf
# File in use by: LENOVO Hostname: DESKTOP-FRVD8ST ProcessID: 16928
# Attempting to use alternate WLF file "./wlfctxj14j3".
** Warning: (vsim-WLF-5001) Could not open WLF file: vsim.wlf
# Using alternate file: ./wlfctxj14j3
VSI7> run -all
# =====
# LFSR_tb Testbench - 10 Test Cases
# =====
# PASS | Test Case 1 | DATA=0x39 -> CRC=0x13
# PASS | Test Case 2 | DATA=0x0c -> CRC=0xba
# PASS | Test Case 3 | DATA=0x8c -> CRC=0x3d
# PASS | Test Case 4 | DATA=0x7d -> CRC=0x1f
# PASS | Test Case 5 | DATA=0x72 -> CRC=0x1e
# PASS | Test Case 6 | DATA=0x47 -> CRC=0xb7
# PASS | Test Case 7 | DATA=0x34 -> CRC=0x76
# PASS | Test Case 8 | DATA=0x2c -> CRC=0xd8
# PASS | Test Case 9 | DATA=0xd8 -> CRC=0x00
# PASS | Test Case 10 | DATA=0x10 -> CRC=0xdc
# =====
# RESULTS: 10 PASSED, 0 FAILED (out of 10)
# =====
# ** Note: $stop : E:/courses/Digital IC design Course/ASIC Flow/My Work/My Assignments/Assignment 5 Verilog/LFSR_tb.v(97)
# Time: 19200 ns Iteration: 0 Instance: /LFSR_tb
# Break in Module LFSR_tb at E:/courses/Digital IC design Course/ASIC Flow/My Work/My Assignments/Assignment 5 Verilog/LFSR_tb.v line 97
```

Figure 2: Simulation Log Output Screen Shot